

Hadisa Abroo

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RESEARCH PROFILE

Researcher in applied and theoretical quantum physics with primary interests in quantum entanglement, quantum coherence, and open quantum systems. Current work focuses on non-Hermitian quantum mechanics and exceptional-point-enhanced sensing, coherent-state quantum metrology under decoherence, coherence-assisted entanglement dynamics in semiconductor quantum dot systems, and photonic quantum platforms, with quantum Fisher information, Schmidt decomposition, and von Neumann entropy as the principal analytical tools. Author of four quantum-theory manuscripts, one sole-authored and three co-authored, and currently undertaking quantum technologies training at the School of Natural Sciences, NUST. Experienced in analytical quantum theory, Lindblad master equation modelling, and quantum circuit simulation via IBM Qiskit.

Keywords Quantum sensing · Quantum metrology · Quantum entanglement

EDUCATION

MPhil in Physics · Riphah International University, Pakistan 2022 – 2025

CGPA 3.38 / 4.0 | Specialisation: Quantum Coherence, Entanglement & Quantum Entropy

MSc in Applied Physics · Riphah International University, Pakistan 2020 – 2022

CGPA 3.53 / 4.0 | Focus: Quantum Mechanics, Electromagnetic Theory, Mathematical Physics

BSc in Mathematics & Physics (Double Major) · University of Sargodha, Pakistan 2017 – 2019

First Division | Core: Classical Mechanics, Optics, Linear Algebra, Calculus

PUBLICATIONS AND PREPRINTS

*Corresponding author. † Manuscript available upon request.

UNDER REVIEW

- [1] **Abroo, H., & Iqbal, A.** * “Quantum Coherence-Assisted Quantum Dot–Photon Entanglement by Schmidt Decomposition Approach.” Submitted to *Physica B: Condensed Matter* (Elsevier), 2026.[†]
- [2] **Abroo, H., Bashir, S., & Iqbal, A.** * “Tunable Photon–Graphene Quantum Dots Entanglement by von Neumann Quantum Entropy.” Submitted to peer-reviewed journal, 2026.[†]

PREPRINTS

- [3] **Abroo, H.** “Exceptional Points in Non-Hermitian Quantum Systems and Their Role in Enhanced Sensing.” Preprint, 2026. (arXiv:quant-ph, submission in progress.)[†]

IN PREPARATION

- [4] **Abroo, H., & Iqbal, S.** * “Quantum Fisher Information of Gazeau–Klauder Coherent States under Markovian Decoherence.” Manuscript in preparation, 2026.

RESEARCH EXPERIENCE AND TRAINING

Quantum Internship 2026 · School of Natural Sciences, NUST, Islamabad Jun – Aug 2026

Supervisor: Prof. Dr. Shahid Iqbal, Department of Physics & Astronomy

- Eight-week in-person programme in quantum technologies, combining lectures with laboratory sessions.
- Quantum circuit design and multi-qubit entanglement generation; quantum communication, cryptography, and teleportation protocols; quantum algorithms and complexity advantage.
- Hands-on implementation in Qiskit and QuTiP; capstone quantum programming project.

Exceptional-Point-Enhanced Quantum Sensing 2025 – 2026

- Analytical treatment of non-Hermitian Hamiltonians; eigenvalue and exceptional-point analysis.
- Sensitivity enhancement characterised via quantum Fisher information; unified analytical framework deriving signal and noise from a single parametrised Hamiltonian.
- PT-symmetry breaking, spectral topology, and connection to quantum metrology protocols.
- Mathematica symbolic simulations; publication-quality figure preparation.

Quantum Coherence-Assisted Entanglement via Quantum Entropy (MPhil Thesis) 2022 – 2025

- Theoretical framework for coherence as a resource for entanglement generation in quantum materials.
- Entanglement quantified by Schmidt decomposition in GaAs/AlGaAs quantum dot systems; entanglement reaches one full ebit at control-laser resonance and is stabilised by increasing control power.
- Density matrix evolution modelled via Lindblad master equation; concurrence, von Neumann entropy, and quantum discord as diagnostic measures.
- IBM Qiskit simulations of entropy-driven entanglement dynamics under decoherence.
- Systematic review of 80+ papers across quantum thermodynamics, quantum communication, and open quantum systems.

Photon–Graphene Quantum Dot Entanglement 2025 – 2026

- Entanglement characterisation in photon–graphene quantum dot hybrid systems.
- Tunable photonic quantum-dot interactions; von Neumann entropy as the primary diagnostic.
- Collaborative theoretical work; manuscript under journal review.

OPEN-SOURCE RESEARCH CODE

Repository: github.com/hadisaabroo99-pixel

PROJECT

	CONTENT	STATUS
Quantum-Enhanced Metrology:	Heisenberg-limit estimation, quantum Fisher information, Ramsey schemes, Ramsey Interferometry, Squeezed States & Standard Quantum Limit	Active
Quantum Entanglement & Nonlocal Correlations:	Bell-state preparation and measurement, CHSH inequality violation, quantum teleportation, entanglement monotones, Bloch sphere visualisation, systematic decoherence simulations. Qiskit + Python. Supports publication [1].	Active
Quantum Computing Algorithms:	Grover's algorithm, surface and repetition code QEC (stabilizer formalism), Grover Search, Quantum Error Correction & Entanglement Scaling	In progress

TECHNICAL SKILLS

Quantum Software	IBM Qiskit, QuTiP, Python (quantum simulation)
Scientific Computing	NumPy, SciPy, Matplotlib, Pandas
Symbolic Computation	Mathematica / Wolfram Language
Scientific Writing	L ^A T _E X (Overleaf), Zotero
Design & Visualisation	Adobe Creative Suite (research figures, posters)

RESEARCH INTERESTS

- Quantum Information Theory
- Quantum Entanglement and Coherence
- Open Quantum Systems and Thermodynamics
- Quantum Sensing and Metrology
- Non-Hermitian Quantum Systems
- Quantum Communication and QKD

SELECTED ACHIEVEMENTS

- Four quantum-theory manuscripts — two submitted to peer-reviewed journals (including *Physica B: Condensed Matter*, Elsevier), one sole-authored preprint, and one in preparation.
- Three open-source quantum simulation repositories on GitHub.
- MSc CGPA 3.53 / 4.0 (top cohort); BSc First Division.
- IELTS Academic Band 7.0 (C1 — English proficiency).

CONFERENCES AND WORKSHOPS

International Conference on Emerging Quantum Sciences & Technologies (ICEQST 2026)	27 Aug 2026
Attendee · School of Natural Sciences, NUST, Islamabad.	

CERTIFICATIONS

Practical Quantum Computing with IBM Qiskit, Packt (UK)	2023
Quantum Mechanics, University of Colorado Boulder (Coursera)	2023
How to Write and Publish a Scientific Paper, École Polytechnique (Coursera)	2022
Scientific Writing with L ^A T _E X & Overleaf, Fred Hutchinson Cancer Center (Coursera)	2022

LANGUAGES

English	C1 — Advanced	IELTS Academic 7.0
German	A1 — Elementary	Actively developing
Urdu	C2 — Native	

ADDITIONAL EXPERIENCE

Research Visualisation Designer · Sixdeees	2021 – present
Scientific poster design, data visualisation, and research communication graphics.	
Physics & Mathematics Instructor · ZA Public Schools / Allied School	2018 – 2020
Secondary-level instruction; curriculum development; science communication.	

REFERENCES

Available upon request. Academic referees from Riphah International University (MPhil thesis supervisor and departmental faculty) and the School of Natural Sciences, NUST.